

TATA CONSULTING ENGINEERS LIMITED		BID ANALYSIS - TECHNICAL	
		Client: GOPIN SEMICON PVT. LTD	REV:-R0
		PROJECT: 3GW SOLAR CELL MANUFACTURING FACILITY AT SURAT, GUJARAT	DATE:- 14-05-2025
Sl.No.	DETAILS	TCE SPECIFICATION	Bidder reply
B	TECHNICAL CHECKLIST - HVAC		
48	THERMAL INSULATION		
1	Application	Ducting	Noted
2	Insulation Material	Class "0" Closed Cell Nitrile Rubber Elastomeric Foam	Comply
3	Lamination	Chemically treated glass cloth (7 mil / 0.18 mm thickness)	Comply
4	Thermal Conductivity	$\leq 0.036 \text{ W/(m-K)}$ at mean temp 24°C (as per ASTM C518 / ASTM C177)	Comply
5	Density	40 to 60 kg/m ³	Comply
6	Water Absorption	$\leq 0.2\%$ by volume (as per ASTM C1763)	Comply
7	Fire Performance Rating	25/50 (as per ASTM E84)	Comply
8	Water Vapor Permeability	≤ 0.05 Perm-inch (as per ASTM E96, Procedure A)	Comply
9	Anti-Microbial Protection	Microban® additive, EPA approved, integral and non-removable	Comply
10	Anti-Microbial Compliance	ASTM G21, UL 181, ASTM E2180	Comply
11	Adhesive	Supplied by principal manufacturer	Comply
			Noted
49	ACCOUSTIC INSULATION		
1	Application	Acoustic insulation for supply air ducts	Noted
2	Insulation Material	Class 1 Open Cell Nitrile Rubber Elastomeric Foam	Comply
3	Insulation Thickness	10 mm	Comply
4	Density	140 – 180 kg/m ³	Comply
5	Fire Rating Standard	BS 476 Part 7	Comply
6	Fire Behavior Compliance	UL 94 compliant (supplementary air distribution materials)	Comply
7	Water Absorption	$\leq 0.2\%$ by volume (as per ASTM C209)	NA
8	Anti-Microbial Compliance	ASTM G21, G22, ASTM E2180 – Should not support microbial growth	Comply
9	Odor Emission	No objectionable odors (as per ASTM C665)	NA
10	Thermal Conductivity	$\leq 0.04 \text{ W/m-K}$ @ 20°C (as per DIN EN 12667)	We have K value 0.0472 W/mk at 20 C
11	Water Vapor Permeance	≤ 0.10	NA
12	Adhesive VOC Compliance	VOC content $\leq 50 \text{ g/L}$	Comply
13	Adhesive Application	Supplied by principal manufacturer	Noted

